LEVEL 17 · PART VIII

GitOps

17

Goal: Operate Kubernetes by declaring desired state in Git and letting a controller continuously reconcile it.

RECOMMENDED TIME
2-3 weeks

TOPICS
7

PRACTICE / CAPSTONE
Install Argo CD (or Flux), deploy the app from a GitOps repo, promote dev → staging → prod via Git, and roll back with git revert.

Topics

#	Topic	What you will learn
1	Desired State Management **p.3305**	GitOps principles, declarative config, pull vs push deploys, repo layout
2	Argo CD and Flux **p.3317**	architecture, install, Applications, Kustomization/HelmRelease, choosing between them
3	Continuous Reconciliation **p.3329**	drift correction, self-heal, prune, sync loops
4	Environment Promotion **p.3341**	overlays, image tag bumps, PR-based promotion, Image Updater/Flux automation
5	Application Synchronization **p.3351**	sync waves, hooks, sync policies, health assessment, ApplicationSets
6	Rollback Through Git **p.3365**	git revert as rollback, audit trail, emergency procedure
7	The Full Flow: CI to GitOps **p.3375**	Developer → Git → CI → Docker image → GitOps repo → Argo CD → Kubernetes, secrets in GitOps (Sealed Secrets/SOPS/ESO)
★	Level capstone project **p.3389**	Install Argo CD (or Flux), deploy the app from a GitOps repo, promote dev → staging → prod via Git, and roll back with git revert.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

